Predicting Used Car Prices: A Practical Machine Learning Approach

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I. Introduction

Valuation of used cars has always been a challenge in the consumer market, as the actual value of a car depends on many technical and sensory factors. Traditional valuation methods often rely on expert judgment or inconsistent market references, leading to a lack of transparency and subjective pricing (Kim & Cho, 2019). While buyers seek fair value, sellers and dealers tend to optimize their profits, resulting in pricing gaps and information asymmetry in the used car market.

This project focuses on building a model to predict used car prices based on historical data from more than 2,000 cars, including 32 input variables reflecting characteristics such as mileage, year of manufacture, vehicle condition, engine capacity, color, number of ownerships, etc. By leveraging algorithms like linear regression, decision trees, random forests, and neural networks, the model is designed to learn complex, nonlinear relationships between features and car prices (Zhou & Jin, 2020).

This issue is key for stakeholders:

- Dealers/manufacturers can refine pricing and forecast inventory.
- Buyers/sellers gain pricing transparency.
- Platforms/analysts improve benchmarking and trend insights (Wang et al., 2020).

The goal is to build a robust model that boosts pricing accuracy, reduces bias, and enhances market efficiency (Kim & Cho, 2019).

II. Data Visualisation

2.1. Dataset Overview

The dataset that was looked at for this study has a lot of information about used vehicles, including several factors that affect their prices. It is set out like a table, with each row showing a different car and each column showing a different feature of that car.

The dataset consists of: 2,230 records, each representing a vehicle. A total of 32 variables, including 1 target variable (vehicle price) and 31 input variables describing specifications, usage history, and additional features. The variables reflect aspects such as engine displacement, mileage, brand, year of manufacture, condition, safety rating, and the presence of specific features such as navigation or cruise control. The price variable, which serves as the prediction target, is continuous and numeric, making this a regression problem.

What makes the dataset particularly well-suited for analysis is the diversity of price points – from entry-level models under $10,000 to luxury vehicles over $40,000. This allows for meaningful exploration of correlations and patterns, and allows machine learning models to be applied across different vehicle segments and price ranges.

The table summarizes several representative variables:

Variable Name	Description	Data Type	Role
Price	Market price of the car (USD)	Continuous numerical	Target variable
Year	Year of manufacture	Integer	Input feature
Brand	Manufacturer brand (e.g., Toyota, Ford)	Categorical	Input feature
Estimated_Mileage	Average monthly mileage	Continuous numerical	Input feature
Engine_Size	Engine displacement in liters	Continuous numerical	Input feature
Condition	Car condition (Excellent, Good, Fair, etc.)	Ordinal categorical	Input feature
Color	Exterior color of the vehicle	Categorical	Input feature
Navigation	Indicates if the car has a navigation system	Binary (Yes/No)	Input feature
Owners	Number of previous owners	Integer	Input feature
Cylinder_Numbers	Number of engine cylinders	Integer	Input feature

Table 1: Sample Variables in the datatset

To better understand the structure of the dataset, each row captures detailed information about an individual used car:

- Example Record 1:

  - Model: Toyota Corolla
  - Year: 2019
  - Fuel Type: Electric
  - Transmission: Manual
  - Number of Doors: 4
  - Condition: Excellent
  - Colour: Beige
  - Cruise Control: Yes
  - Leather Seats: No
  - Heated Seats: Yes
  - Navigation: Yes
  - Owners: 0
  - Wheel Type: Chrome
  - Sunroof: Yes
  - Warranty: 35 months
  - Age: New
  - Price: $25,385.24
  - Cylinder Numbers: Four
  - Credit History: 0.006731
  - Engine Size: 2.70 liters
  - Weight: 3543.01 lbs
  - Monthly Mileage: 894.74 miles
  - Peak RPM: 4689
  - Estimated Mileage: 10,736.91 miles

- Example Record 2:

  - Model: Nissan Altima
  - Year: 2007
  - Fuel Type: Petrol
  - Transmission: Manual
  - Doors: 4
  - Condition: Fair
  - Color: Brown
  - Cruise Control: No
  - Leather Seats: No
  - Heated Seats: No
  - Navigation: No
  - Owners: 2
  - Insurance: Third Party
  - Service History: No Service
  - Safety: Not rated
  - Wheel: Alloy
  - Sunroof: No
  - Warranty: 7
  - Price: $8,005.47
  - Cylinder Numbers: Two
  - Credit History: 0.004967
  - Engine Size: 1.66 L
  - Weight: 3,170 lbs
  - Car Length: 179.1 inches
  - Car Width: 68.1 inches
  - Monthly Mileage: 3,020 miles
  - Estimated Mileage: 36,246 miles
  - Peak RPM: 6000
  - Estimated Mileage: 36,246 miles

In summary, this dataset provides a well-rounded basis for developing predictive models, offering both the complexity and structure needed for effective data preprocessing, feature selection, and regression analysis.

2.2. Variable Selection

In this project, feature importance was assessed using a Random Forest Regressor.

After removing the Monthly_Mileage variable due to its strong correlation and content overlap with Estimated_Mileage, the Random Forest model was trained with n_estimators = 100 and random_state = 42. A new variable, Car_Age, was also created by subtracting the manufacturing year from the current year to capture the age of the vehicle. The dataset was split into 70% for training and 30% for testing. The model's feature_importances_ attribute was then used to extract the contribution score of each variable.

The results identified the top 10 features that contributed most significantly to price prediction. These include engine size, weight, and number of cylinders; usage-related

factors such as ownership history and mileage; as well as descriptive attributes such as brand, color, and safety.

Variable Name	Description	Data Type	Importance Score
Estimated_Mileage	Average monthly mileage	Continuos numerical	0.9360
Color	Exterior color of the vehicle	Categorical	0.0113
Model	The brand and model of the car	Categorical	0.0091
Owners	Number of previous owners	Integer	0.0053
Engine_Size	The size of car's engine	Continuos numerical	0.0051
Safety	The safety rating of the car	Ordinal categorical	0.0044
Weight	The weight of the car	Continuous numerical	0.0033
Service_History	Information about the car's service history	Categorical	0.023
Insurance	The type of insurance coverage for the car	Categorical (binary)	0.023
Car_Age	The current year minus the year the car was manufactured.	Integer	0.023

Table 2: Top 10 Selected Features

2.3. Data Cleaning

The dataset was divided into two groups: categorical and numerical variables. For the numerical columns, raw values often included units or formatting characters (e.g., "km", commas, periods, dashes). These were stripped or replaced with null values (NaN) before converting the fields to floats using pd.to_numeric.

Several corrections were applied:

- In TAge, the invalid value 50 was replaced with 5.
- For Owners, outliers such as 22 and 44 were normalized to 2 and 4.
- In DoorsNum, values like 20 and 40 were mapped to 2 and 4 respectively.

For the categorical columns, inconsistencies in casing, spelling, and extra whitespace were corrected:

- Fuel_Type: "Petroll" was corrected to "Petrol".
- Transmission: "AutoNamatic" was fixed to "Automatic".
- Condition: entries like "Exc" and "Excellent!" were unified under "Excellent".
- Color: misspelled values such as "Brwn" were changed to "Brown".
- Insurance: "Y" was replaced with "Yes".
- Service_History: variants of "Partial" and "Full" were consolidated.
- Safety: numeric values were converted to string labels (e.g., "4" → "4 stars").
- Cylinder_Numbers: numeric values were mapped to words (e.g., "3" → "three").

After standardization, all categorical variables were label-encoded to make them compatible with machine learning models.

2.4. Univariate Analysis

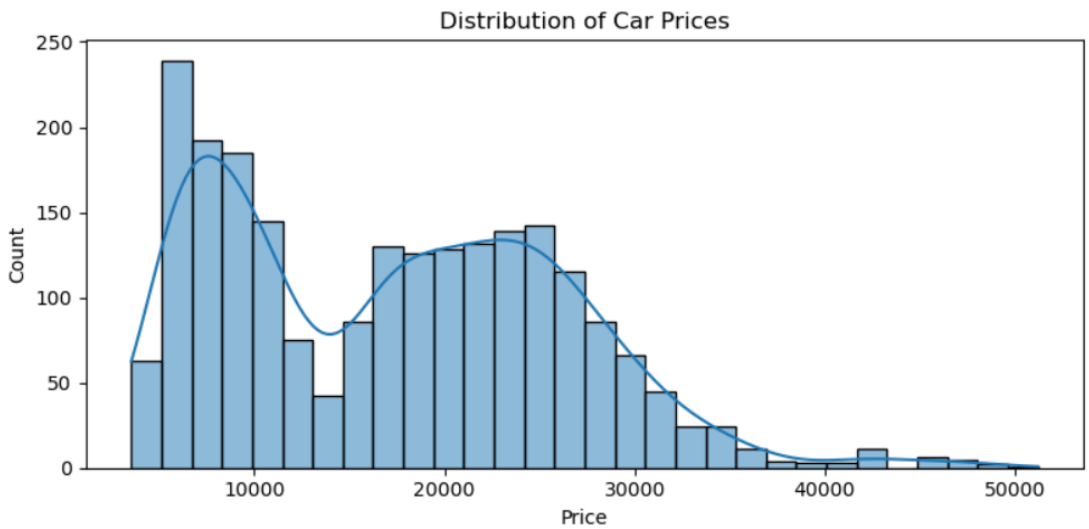

Figure 1:Distribution of Car Prices

Price is right-skewed, with most cars under $25,000 and a few high-end outliers above $40,000. The multi-modal shape suggests distinct segments (economy, mid-range, high-end), with $7,000–$10,000 being the most common range.

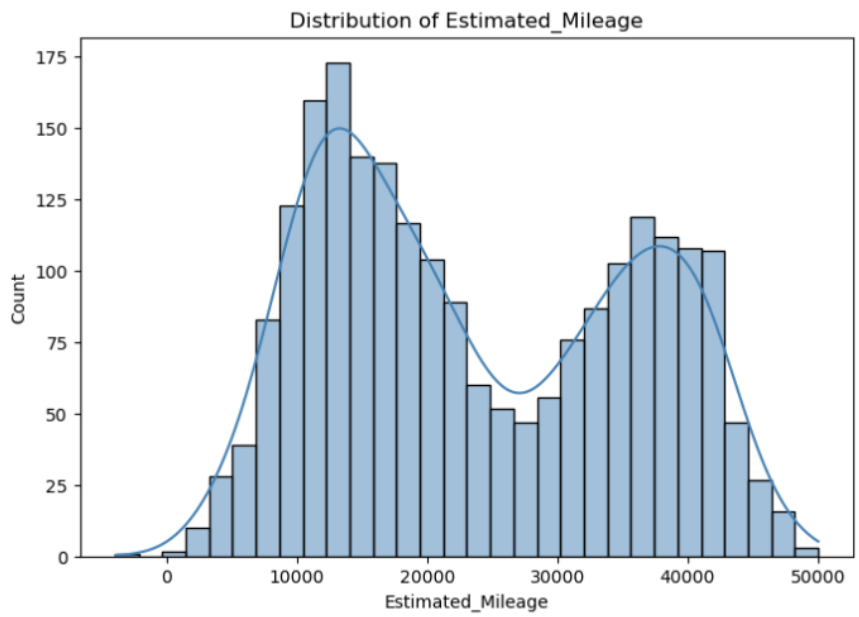

Figure 2: Distribution of Estimated Mileage

Estimated_Mileage displays a bimodal distribution, with two pronounced peaks around 15,000 and 38,000 miles. This suggests a natural split between low- and high-mileage vehicles.

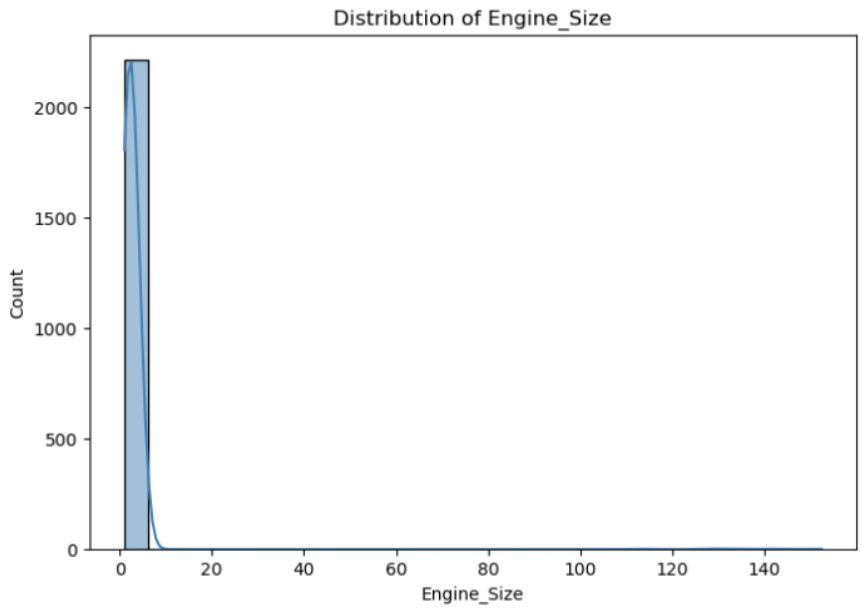

Figure 3: Distribution of Engine Size

Engine_Size is highly right-skewed, with the vast majority of observations clustered under 10 liters.

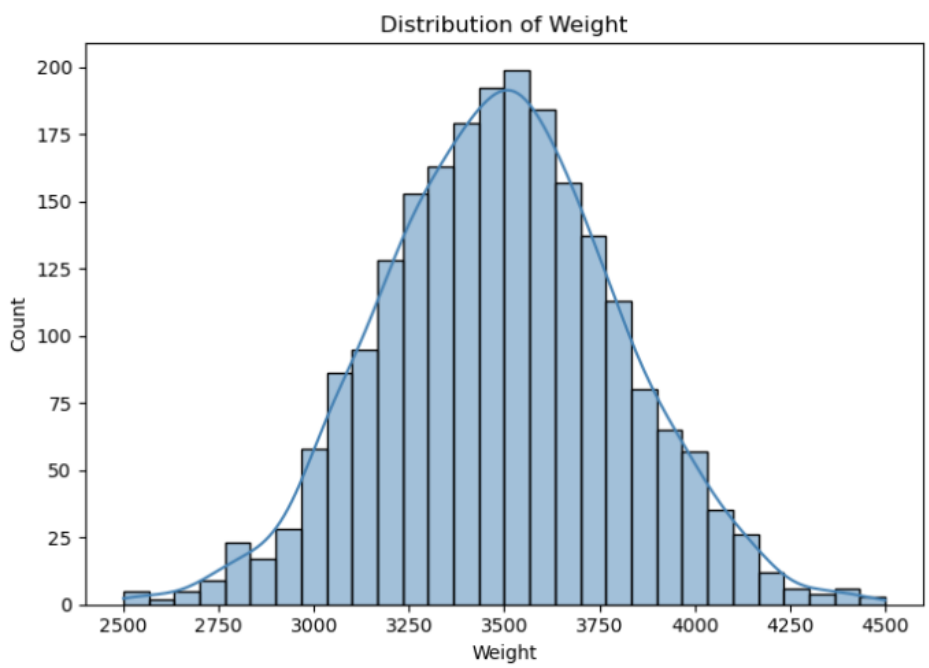

Figure 4: Distribution of Weight

Weight follows a near-normal distribution, centered around 3,500 lbs. Its symmetric bell-shaped curve makes it a stable and reliable variable for modeling without requiring transformation.

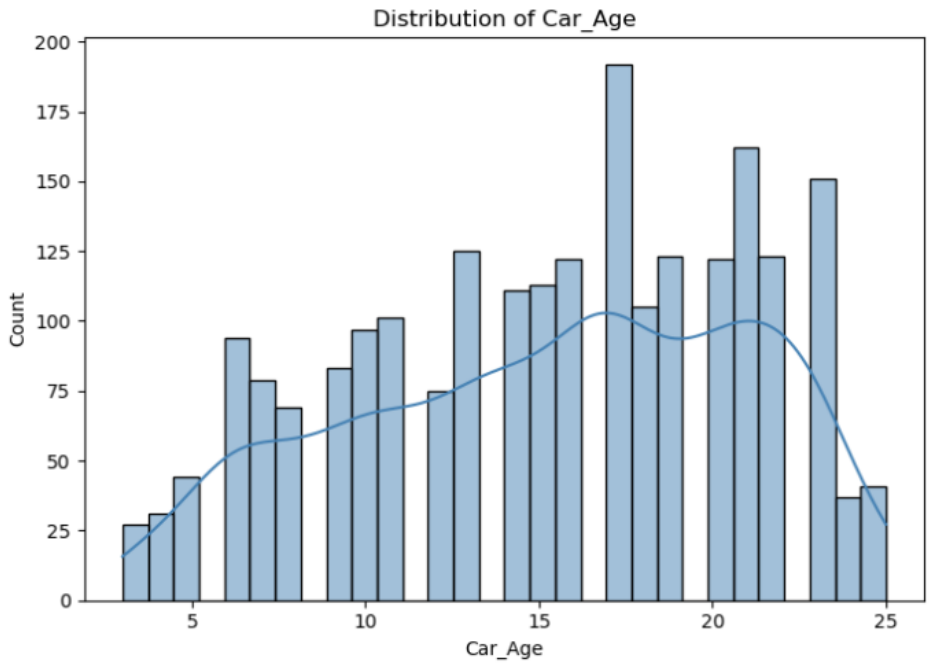

Figure 5:Distribution of Car Age

Car_Age spans a wide range, from approximately 3 to 25 years. The distribution is relatively uniform, reflecting a diverse dataset in terms of vehicle age and usage history.

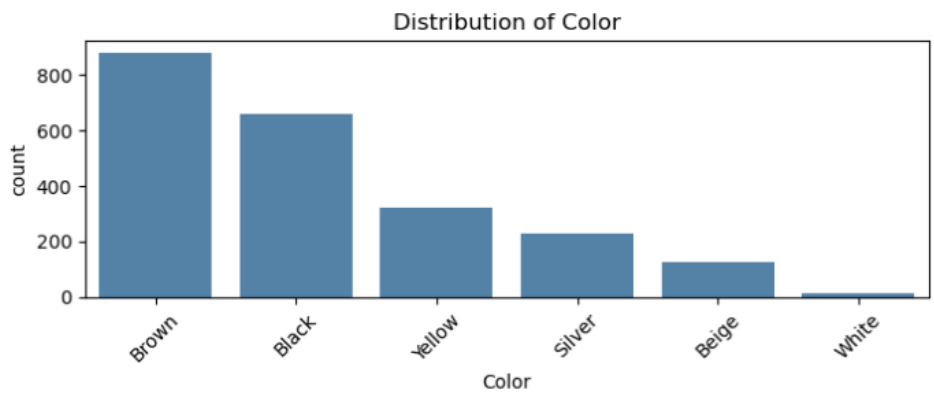

Figure 6: Distribution of Color

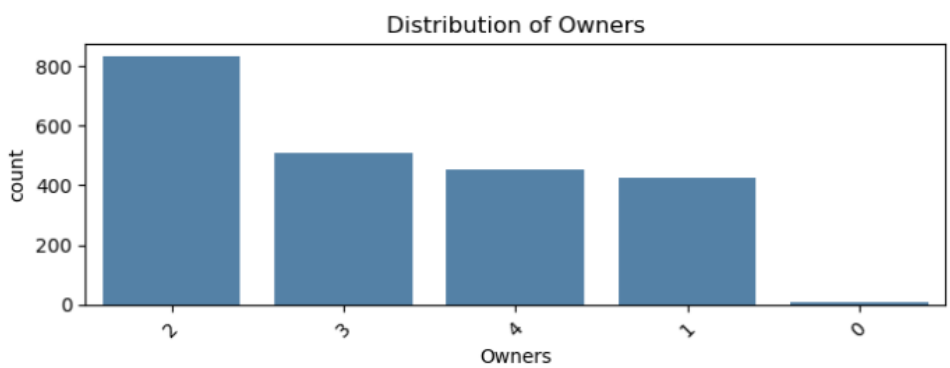

Figure 7: Distribution of Owners

Among categorical variables, Color is dominated by brown, followed by black and yellow. Lighter colors such as white and beige are underrepresented. Owners tends to cluster around 2–3 previous owners, while vehicles with 0 or more than 5 owners are relatively rare.

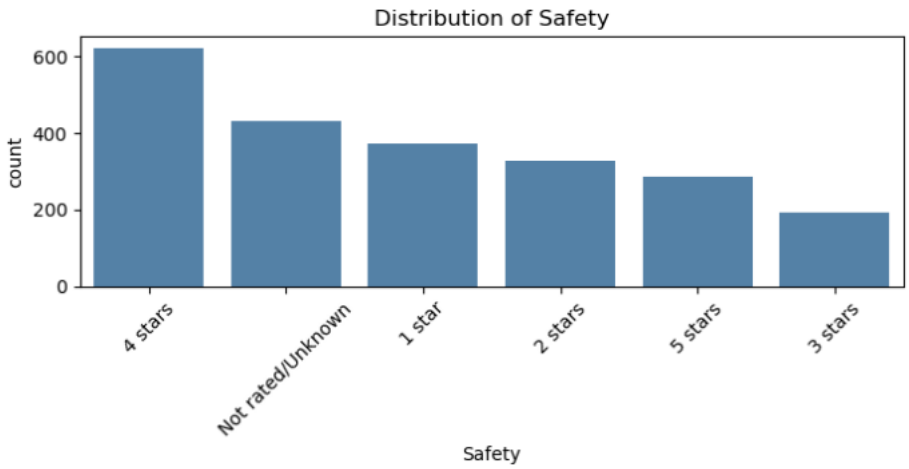

Figure 8: Distribution of Safety

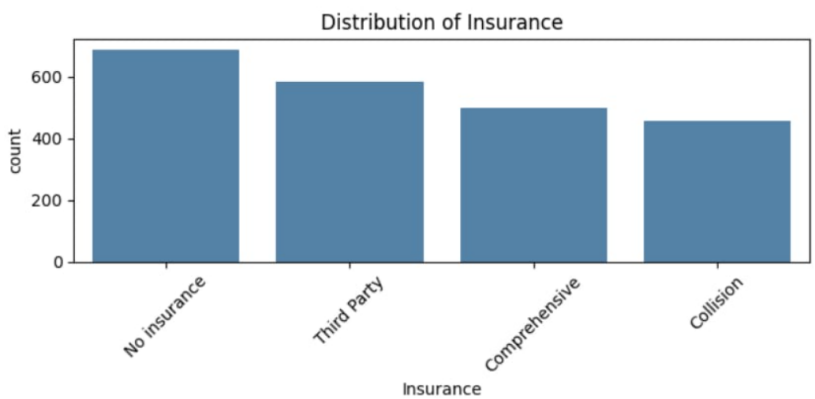

Figure 9: Distribution of Insurance

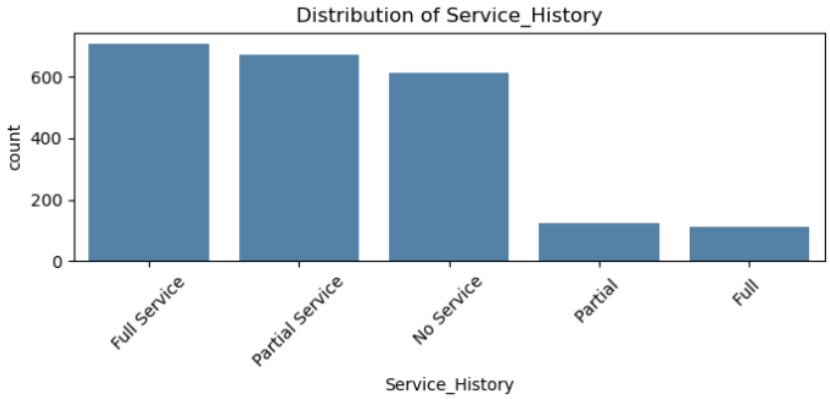

Figure 10: Distribution of Service History

Safety ratings are concentrated at “4 stars” or “Not rated,” with lower ratings spread more evenly. In the Insurance variable, common types such as “Comprehensive,” “Third Party,” and “Collision” are well represented, although a notable portion of vehicles have no insurance coverage. Service_History is mostly complete, with “Full” and “Partial” coverage dominating the distribution, whereas “No service” and “Partial” are less frequent.

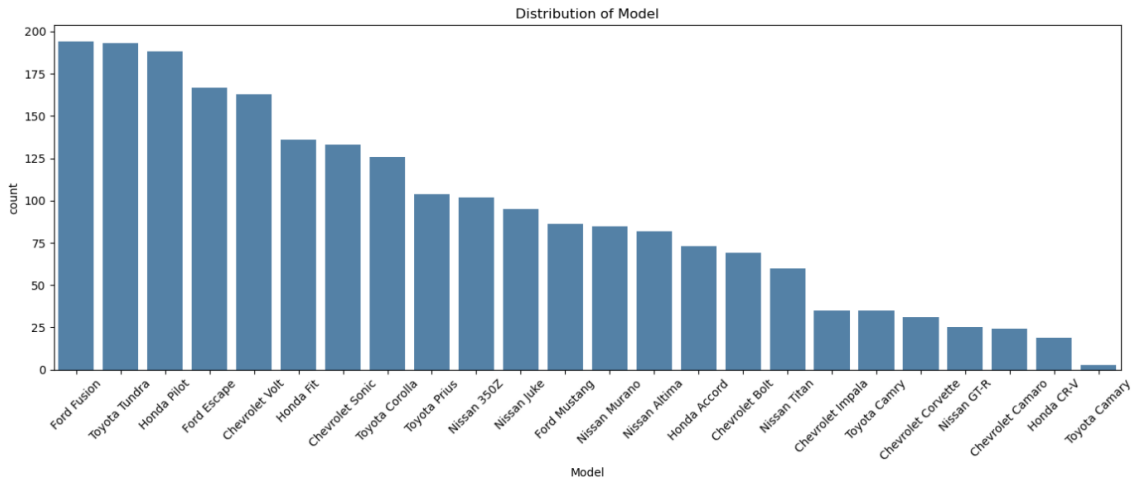

Figure 11: Distribution of Model

The Model variable is relatively balanced across vehicle types. However, Ford Fusion, Toyota Tundra, and Honda Pilot are the most common models in the dataset, each appearing in over 180 records.

2.5. Bivariate Analysis

The relationship between the target variable (Price) and key predictors was examined using scatter plots and boxplots.

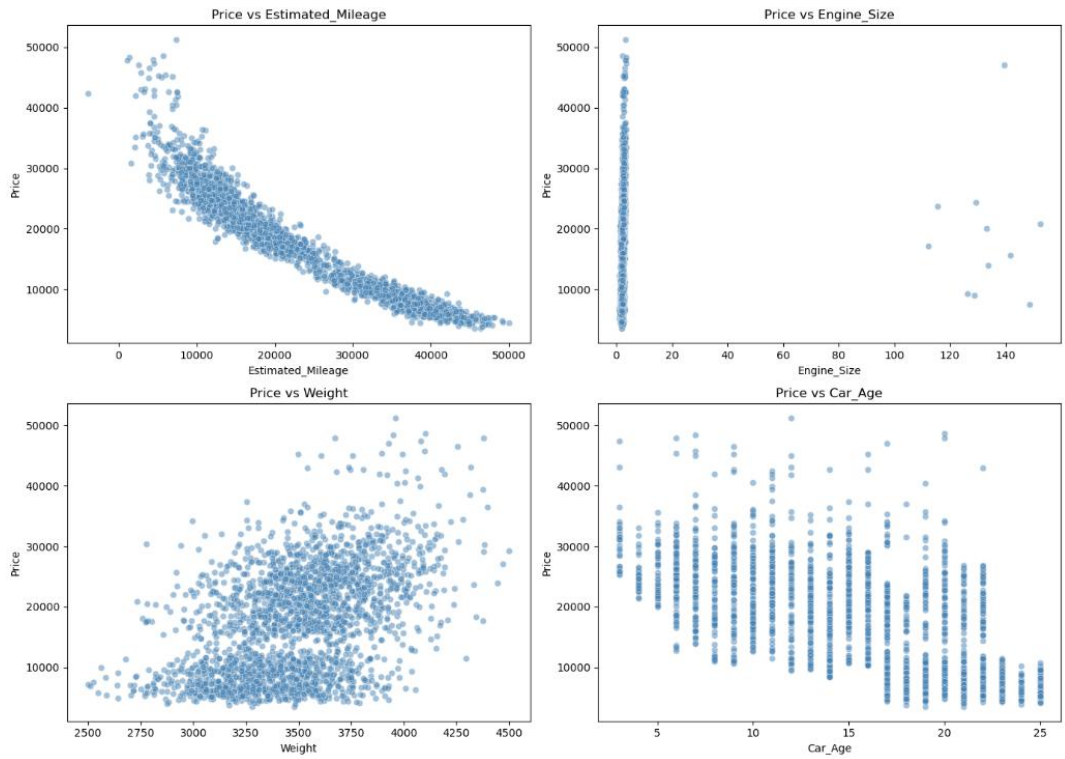

Figure 12: Relationship between Price and numerical variables

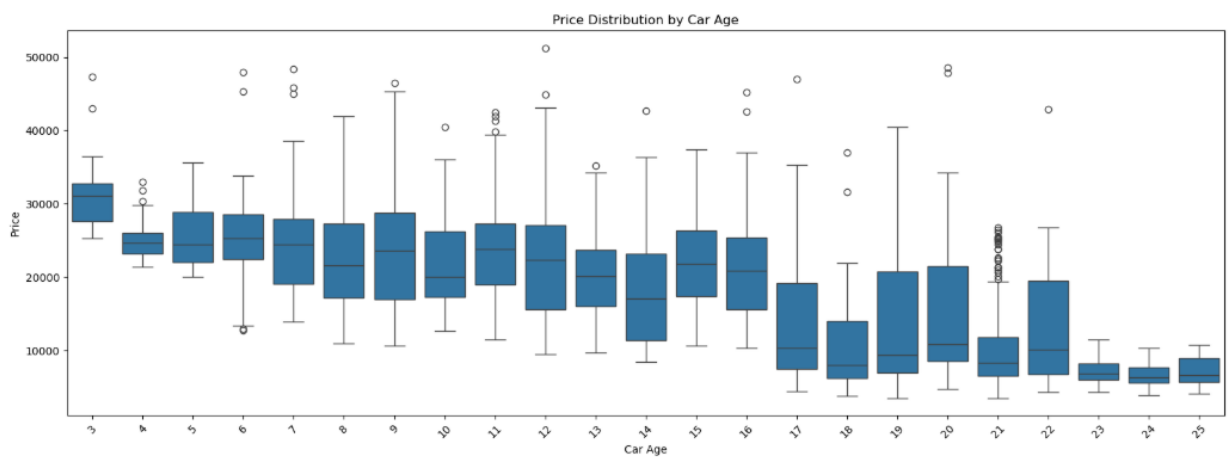

Figure 13: Price Distribution by Car Age

Numerical variables:

- **Estimated_Mileage** shows a strong negative correlation with price. Cars with higher mileage tend to be significantly cheaper, making this the most predictive feature in the dataset.
- **Engine_Size** is highly concentrated under 10, but several extreme values over 100 distort the pattern. While larger engines generally associate with higher prices, the effect is muted due to outliers.

- **Weight** shows a weak positive trend, with heavier vehicles slightly more expensive. However, the large spread suggests that its predictive strength is limited.
- **Car_Age** displays a clear downward trend: newer cars (especially those aged 3–7) hold higher and more stable prices, while vehicles older than 17 years show both lower and compressed price ranges. This makes car age a strong and non-linear predictor of price

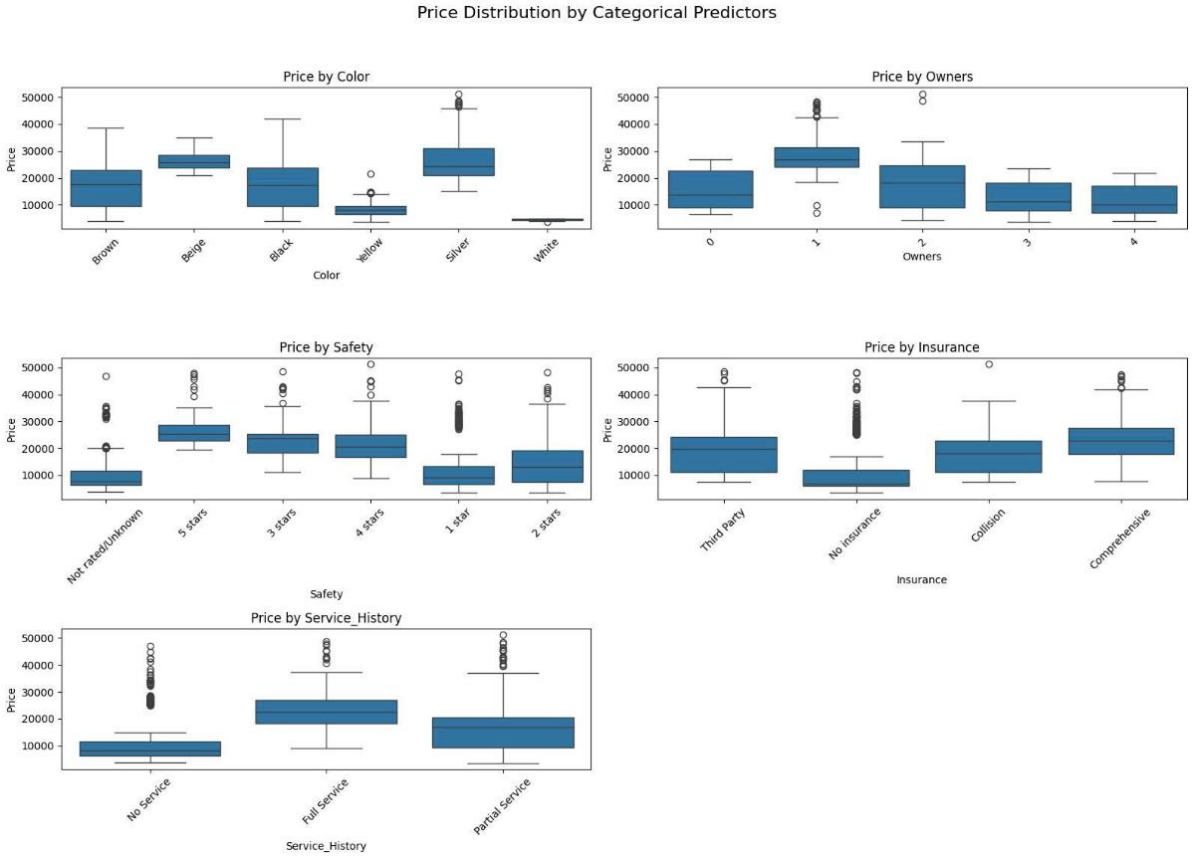

Figure 14: Relationship between Price and Categorical variables

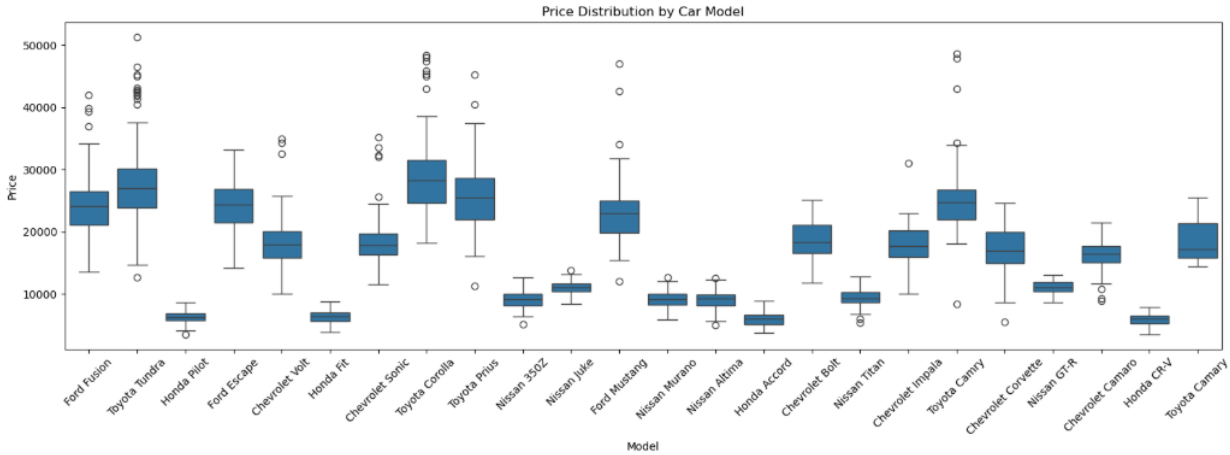

Figure 15: Price Distribution by Car Model

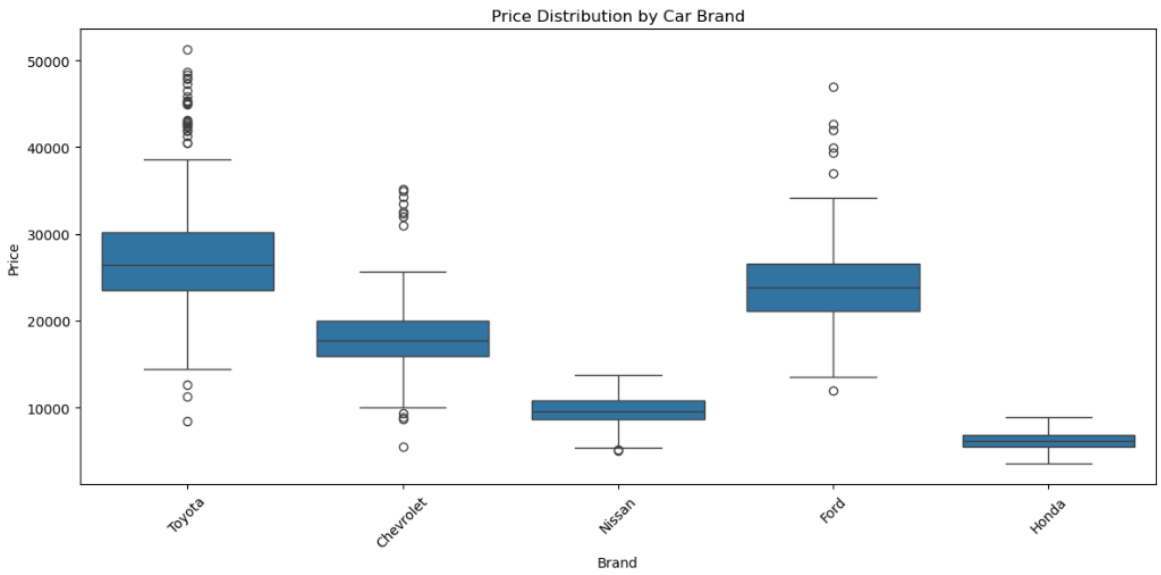

Figure 16: Price Distribution by Car Brand

Categorical variables:

- **Color** does not show a consistent price pattern. Silver and beige cars tend to have slightly higher median prices, but overall, color appears to have low predictive value.
- **Owners** reveals a logical trend: prices peak for cars with one previous owner and decrease steadily as the number of owners increases, confirming moderate predictive potential.
- **Safety** ratings correlate moderately with price. Higher-rated cars (e.g., 5 or 3 stars) are priced higher, while unrated vehicles have the lowest medians, suggesting that safety perception impacts value.
- **Insurance** and **Service_History** are highly informative. Cars with comprehensive insurance or full service records consistently show higher prices, reinforcing their relevance for prediction.
- **Model** exhibits wide price variation across individual car types. When models are grouped into broader **Brand** categories, patterns become clearer. Toyota and Ford lead in median price and range, while Honda and Nissan sit at the lower end. Grouping models by brand simplifies interpretation and improves model clarity.

Overall, variables such as mileage, car age, service history, and brand provide the most explanatory power, while others like weight or color contribute less to price variation.

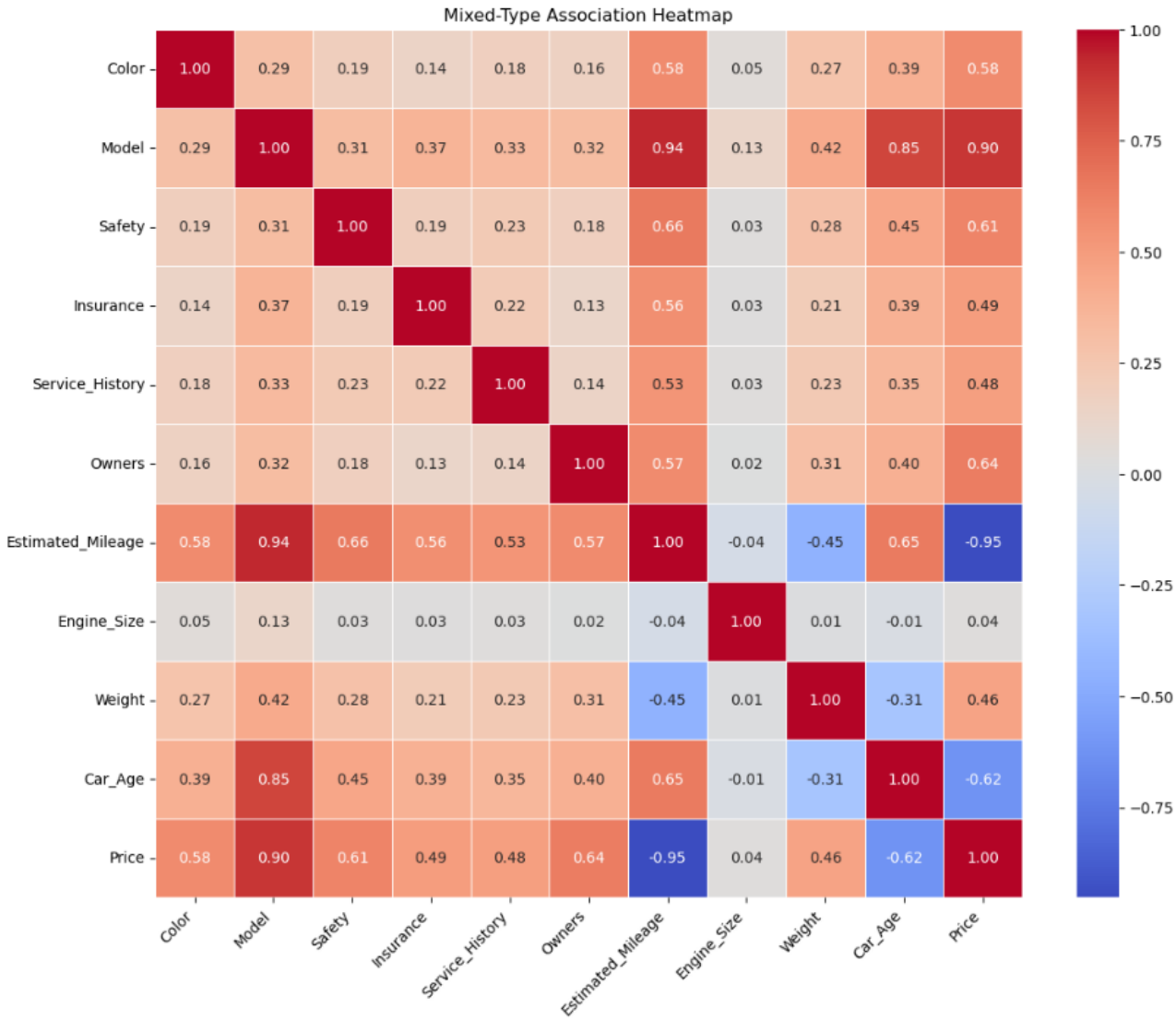

Figure 17: Heatmap correlations

The heatmap reveals three main predictors of price: Estimated_Mileage (–0.95), Model (0.90), and Car_Age (–0.62). Higher mileage and age reduce price, while Model is positively linked to it. Strong internal correlations—Model–Mileage (0.94), Model–Age (0.85), and Age–Mileage (0.65)—suggest older models tend to have more mileage. Other variables like Color and Insurance show minimal impact. Overall, usage, model type, and age are key drivers of price.

2.6. Multivariate Analysis

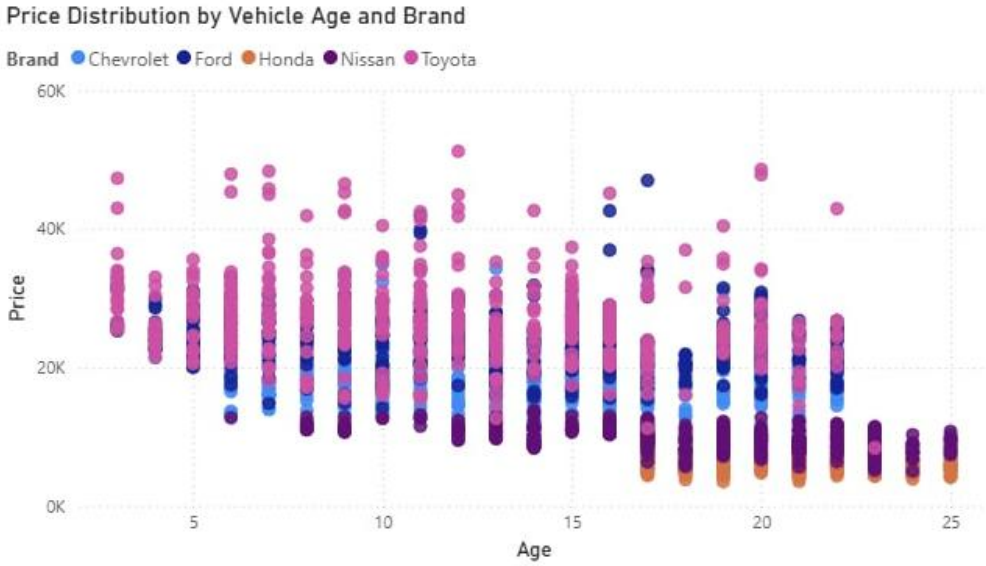

Figure 18: Price Distribution by Vehicle Age and Brand

The scatter plot illustrates how vehicle price varies jointly with age and brand. Price declines with age across all brands, with the steepest drop in the early years. Toyota and Ford consistently appear in the higher price tiers, while Honda and Nissan cluster in the lower segments. Some outlier Ford and Chevrolet vehicles maintain elevated prices even at older ages—likely classic or special-edition models.

2.7. Data Quality Assessment and Treatment

Some features contained missing data, specifically Color (2 records), Safety (430 records), and Weight. For categorical variables like Color and Safety, missing values were filled using the mode (the most frequent category). For the numerical variable Weight, missing values were replaced with the median to reduce the impact of potential outliers.

Before training the model, all categorical variables were encoded into numerical format using Label Encoding. This process assigned a unique integer to each category in selected features such as Model, Safety, Insurance, Service_History, and Color. Label encoding was chosen for its simplicity and compatibility with tree-based models.

After encoding, the dataset was cleaned by removing potential outliers. The Isolation Forest algorithm was applied to all features except the target variable (Price) to detect abnormal observations. A contamination rate of 3% was used to estimate the proportion of outliers. Observations classified as outliers (label = –1) were removed from the dataset.

- ✓ Initial observations: 2,230
- ✓ After removing outliers: 2,163
- ✓ Records removed: 67

These two steps helped standardize the dataset and improve the overall reliability and generalization of the model.

III. Model and Results

3.1. Problem Formulation

This project use structured data containing things like mileage, engine size, model, age, and service history to estimate how much cars will cost. The task is set up as a regression problem since the target variable is continuous. We use two types of models: tree-based models and linear regression.

3.2. Data Partitioning

To assess model performance on unseen data, we split the dataset into 80% training and 20% testing using a fixed random seed (random_state=5) for reproducibility and consistent results.

3.3. Performance Metrics

To evaluate the model's predictive accuracy, we selected Mean Squared Error (MSE) and Root Mean Squared Error (RMSE) as the primary performance metrics.

- Mean Squared Error (MSE) calculates the average of the squared differences between predicted and actual values. It penalizes larger errors more heavily, making it useful when significant deviations are particularly undesirable.
- Root Mean Squared Error (RMSE) is the square root of MSE. It retains the same unit as the target variable (car price), which makes it easier to interpret in a real-world context.

These metrics were chosen because they are well-suited for regression tasks and provide clear insights into the model’s accuracy. Lower MSE and RMSE values indicate better model performance. Additionally, RMSE is especially helpful for understanding how far off predictions are from actual prices on average, making it particularly interpretable in real-world applications (Chai & Draxler, 2014; Hyndman & Koehler, 2006).

3.4. Baseline Model

We used a Dummy Regressor with the "mean" strategy as a baseline, predicting the average training price for all instances. It performed poorly:

The model did quite poorly after training:

- MSE (Training): 75,744,483.75
- RMSE (Training): 8703.13

These high errors show that predicting the mean is ineffective. This baseline highlights the need for more advanced models that leverage feature relationships to improve accuracy.

3.5. Model Training & Improvement

In this project, supervised learning algorithms were used to predict car prices based on input features.

3.5.1. Linear Regression

The Linear Regression model, trained on one-hot encoded categorical and numerical features, shows clear improvement over the baseline:

```
Training data MSE = 4206919.5509613175
Testing data MSE = 5870532.777982672
Training data RMSE = 2051.0776560045983
Testing data RMSE = 2422.9182359259817
```

Figure 19: Linear Regression Model Results

It achieves lower MSE and RMSE on both training and testing sets. However, the ~18% increase in test RMSE suggests mild overfitting. While it captures main trends, the RMSE (~2,400) remains high, indicating that linear regression may not fully model the complexity in car pricing.

3.5.2. Decision Tree

To evaluate the Decision Tree model, six different models were trained using various combinations of max_depth and min_samples_split.

	Model	Params	MSE Train	MSE Test	RMSE Train	RMSE Test
2	Model 3	{'max_depth': 5, 'min_samples_split': 10}	2.93952e+06	4.78302e+06	1714.5	2187.01
0	Model 1	{'max_depth': 5, 'min_samples_split': 20}	2.94245e+06	4.78726e+06	1715.36	2187.98
1	Model 2	{'max_depth': 10, 'min_samples_split': 20}	1.77494e+06	5.09112e+06	1332.27	2256.35
5	Model 6	{'max_depth': 15, 'min_samples_split': 20}	1.52056e+06	5.22889e+06	1233.11	2286.68
3	Model 4	{'max_depth': 10, 'min_samples_split': 10}	1.27973e+06	5.82512e+06	1131.25	2413.53
4	Model 5	{'max_depth': 15, 'min_samples_split': 10}	818711	6.06746e+06	904.83	2463.22

Figure 20: Decision Tree Model Evaluation (with different parameters)

Among six configurations, Model 3 (max_depth = 10, min_samples_split = 10) gave the best test RMSE (2187.01), outperforming both the baseline and linear model. In contrast, Model 6 (max_depth = 15) achieved the lowest training error but had the highest test RMSE (2463.22), suggesting overfitting. These results highlight the trade-off between model complexity and generalization, and the importance of tuning depth and split thresholds to avoid high variance.

3.5.3. Random Forest

Ensemble learning was introduced using Random Forests, which showed significant improvement after tuning tree depth and number of estimators. The best-performing configuration was Model 7 with

	Model	Params	MSE Train	MSE Test	RMSE Train	RMSE Test
6	Model 7	{'n_estimators': 100, 'max_depth': 10, 'min_samples_split': 10}	1.13076e+06	4.00862e+06	1063.37	2002.15
7	Model 8	{'n_estimators': 50, 'max_depth': 10, 'min_samples_split': 10}	1.16179e+06	4.01454e+06	1077.86	2003.63
8	Model 9	{'n_estimators': 30, 'max_depth': 10, 'min_samples_split': 10}	1.18402e+06	4.03456e+06	1088.13	2008.62
5	Model 6	{'n_estimators': 30, 'max_depth': 10, 'min_samples_split': 20}	1.6288e+06	4.14397e+06	1276.24	2035.67
4	Model 5	{'n_estimators': 50, 'max_depth': 10, 'min_samples_split': 20}	1.63024e+06	4.15227e+06	1276.81	2037.71
3	Model 4	{'n_estimators': 100, 'max_depth': 10, 'min_samples_split': 20}	1.61327e+06	4.16073e+06	1270.15	2039.79
2	Model 3	{'n_estimators': 30, 'max_depth': 5, 'min_samples_split': 20}	2.57262e+06	4.51784e+06	1603.94	2125.52
1	Model 2	{'n_estimators': 50, 'max_depth': 5, 'min_samples_split': 20}	2.57242e+06	4.54894e+06	1603.88	2132.82
0	Model 1	{'n_estimators': 100, 'max_depth': 5, 'min_samples_split': 20}	2.56902e+06	4.62391e+06	1602.82	2150.33

Figure 21: Random Forest Model Evaluation (with different parameters)

Compared to the baseline and linear regression models, Random Forest significantly improved test RMSE. The results also suggest that increasing the number of estimators helps the model learn more stable patterns, while moderate tree depth prevents overfitting.

3.5.4. Catboost

CatBoost was used as a gradient boosting solution that handles categorical variables natively, offering competitive results with tuned depth and iterations.

	Model	Params	MSE Train	MSE Test	RMSE Train	RMSE Test
2	Model 3	{'iterations': 500, 'depth': 5}	876433	4.01673e+06	936.18	2004.18
0	Model 1	{'iterations': 300, 'depth': 5}	948157	4.04266e+06	973.73	2010.64
6	Model 7	{'iterations': 500, 'depth': 10}	69414.1	4.09656e+06	263.47	2024
1	Model 2	{'iterations': 400, 'depth': 5}	912604	4.16098e+06	955.3	2039.85
3	Model 4	{'iterations': 600, 'depth': 5}	846354	4.16207e+06	919.98	2040.11
5	Model 6	{'iterations': 400, 'depth': 10}	68832.1	4.29508e+06	262.36	2072.46
7	Model 8	{'iterations': 600, 'depth': 10}	59222.4	4.36118e+06	243.36	2088.34
4	Model 5	{'iterations': 300, 'depth': 10}	94667.8	4.40302e+06	307.68	2098.34
9	Model 10	{'iterations': 400, 'depth': 15}	1416.65	4.87105e+06	37.64	2207.04
10	Model 11	{'iterations': 500, 'depth': 15}	1129.48	4.89696e+06	33.61	2212.91
8	Model 9	{'iterations': 300, 'depth': 15}	1898.86	4.95143e+06	43.58	2225.18
11	Model 12	{'iterations': 600, 'depth': 15}	1010.8	5.05421e+06	31.79	2248.16

Figure 22: Catboost Model Evaluation (with different parameters)

Among all configurations, Model 3 (iterations = 500, depth = 5) performed best with the lowest test RMSE of 2004.18, and a training RMSE of 936.18. This result is comparable to the best-performing Random Forest models, confirming that CatBoost is effective on this dataset.

However, deeper models like Model 12 (depth = 15) achieved extremely low training RMSE (31.79) but much higher test RMSE (2248.16), indicating strong overfitting. This shows CatBoost's power to learn complex patterns, but also highlights the need for careful regularization.

3.5.5. Bagging Neural Network

To test ensemble learning on neural networks, bagging was applied to MLP models, but performance still lagged behind tree-based methods.

```
Train MSE: 8272136.60
Test MSE: 9529745.98
Train RMSE: 2876.13
Test RMSE: 3087.03
```

Figure 23: Bagging Neural Network Results

The Bagging Neural Network model showed slight improvement over a standalone neural network. With a test RMSE of 3,087.03, its performance lags behind Random Forest and CatBoost, which achieved lower errors with more stability.

3.6. Best Model Summary

Throughout the project, we followed a step-by-step improvement strategy to enhance model performance. Starting from a baseline model using mean prediction, we gradually applied more complex techniques, including linear regression with one-hot encoding, decision trees with hyperparameter tuning, and ensemble models like Random Forest and CatBoost. The experiments demonstrate a deliberate effort to balance bias and variance and identify the most suitable approach for car price prediction.

Model	Key Settings	RMSE Train	RMSE Test
Dummy Regressor	Mean prediction only	8703.13	8703.13
Linear Regression	One-hot encoding	2051.08	2422.91
Decision Tree (Model 3)	max_depth=10, min_samples_split=10	1714.50	2187.01
Random Forest (Best)	n_estimators=100, max_depth=10, min_split=10	1063.37	**2002.15**
CatBoost (Model 3)	iterations=500, depth=5	936.18	2004.18
Bagging Neural Network	MLP(hidden_layer=100), n_estimators=10	2876.13	3087.03

Table 3: Model Comparison Table with Key Settings and RMSE Scores

Among all models, Random Forest achieved the best balance between training and test RMSE, with the lowest test error while maintaining strong generalization. Although CatBoost produced a slightly lower training RMSE, it did not outperform Random Forest on the test set. Neural network models underperformed, likely due to limited tuning and higher sensitivity to input scaling.

The final model selection reflects a structured improvement process, where each algorithm was evaluated and optimized to identify the most effective solution for predicting used car prices.

3.7. Error Cost Analysis

This project involves multiple stakeholders:

- Used car sellers, who rely on accurate price predictions to list vehicles competitively.
- Buyers, who use the platform's estimates to assess value and fairness.
- Online platforms, which aim to maintain user trust and encourage successful transactions.

Prediction errors can be categorized as:

Overestimation (predicted price > actual price):

→ Sellers face longer listing durations and fewer inquiries.

→ Buyers might avoid engagement or perceive the platform as unreliable.

Underestimation (predicted price < actual price):

→ Sellers risk accepting lower offers and losing potential profit.

→ Platforms may lose seller confidence and damage brand reputation.

Although both errors are undesirable, underestimation poses the higher cost. It results in direct financial loss for sellers and undermines trust in automated systems. For platforms aiming to scale automated valuation tools, avoiding large underpredictions is critical.

Therefore, error mitigation efforts (e.g., tuning models to reduce left-tail residuals) should prioritize minimizing underestimation, especially for high-value vehicles.

IV. Conclusion

This project aimed to develop a system that helps estimate used car prices based on available information like model, age, mileage, and features. By testing different methods, we found that models using combined decision-making techniques (like Random Forest and CatBoost) gave the most accurate and stable predictions.

While the results are promising, there's still room for improvement. For example, using more detailed vehicle information such as service records, accident history, or regional market trends could make future predictions even more reliable.

As a next step, we suggest integrating this pricing tool into your platform to assist both sellers and buyers. It can improve pricing transparency, reduce manual effort, and help build trust.

In the future, the same approach can be applied to similar products like motorbikes or electric vehicles, making the platform more versatile and competitive.

V. References

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